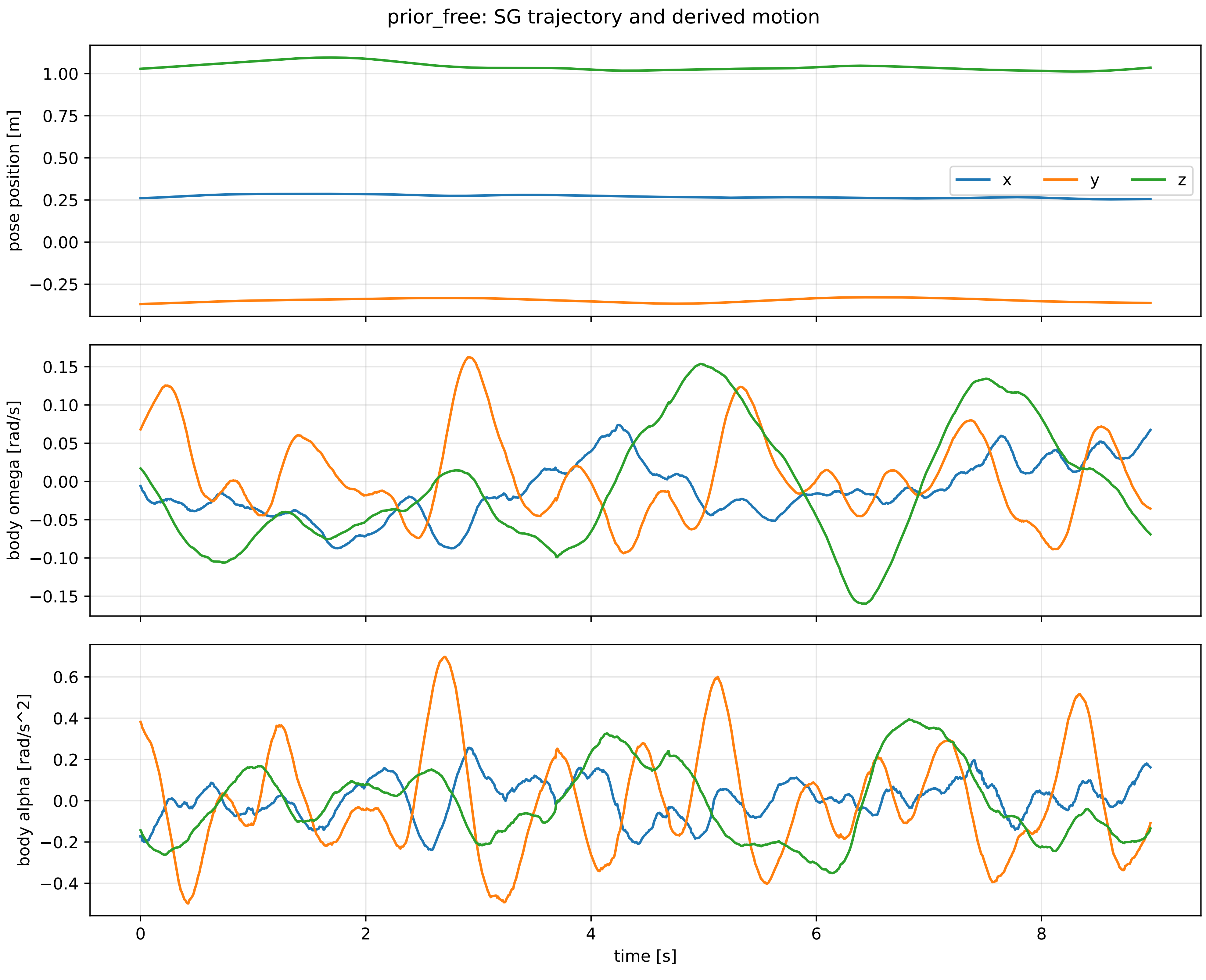
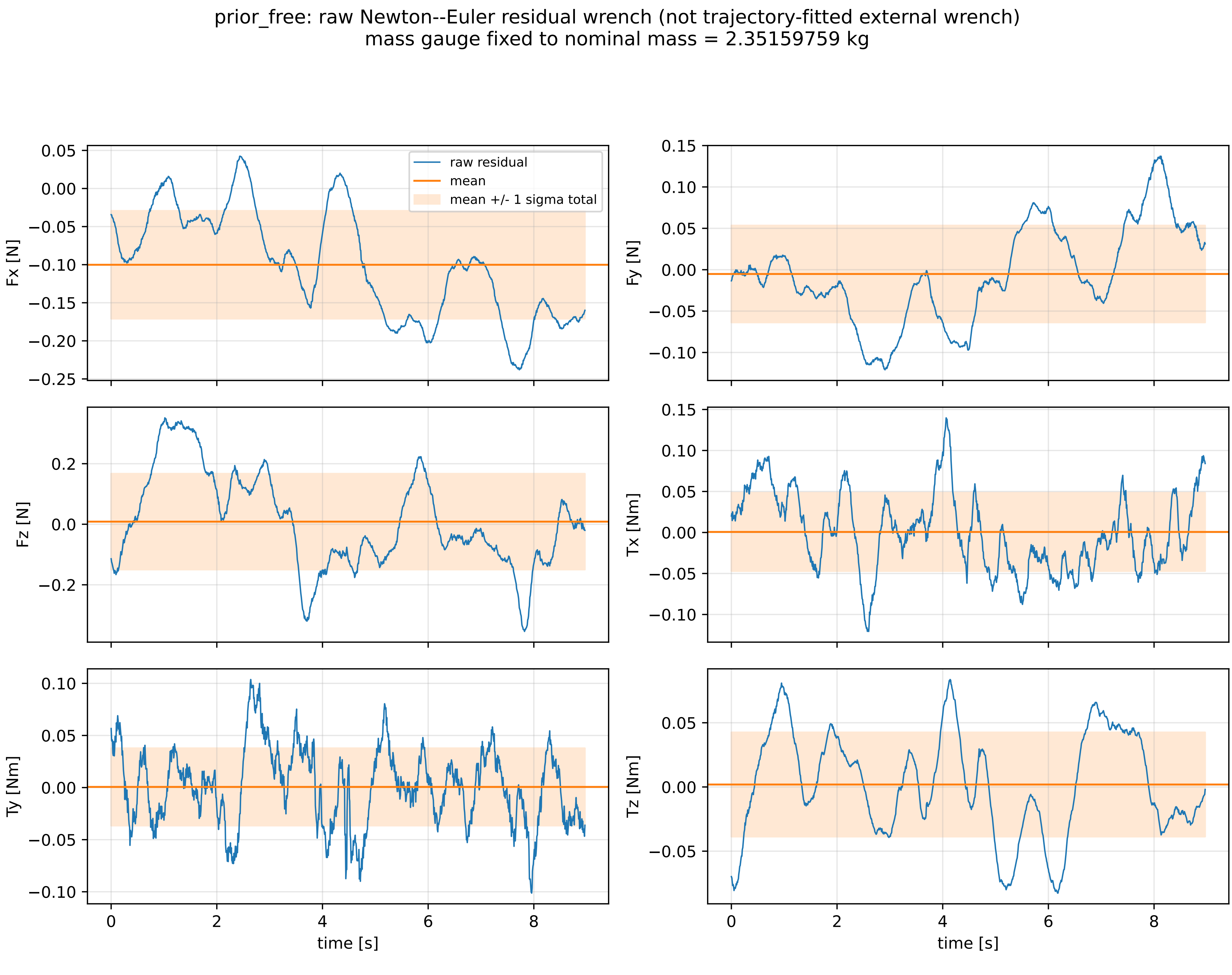
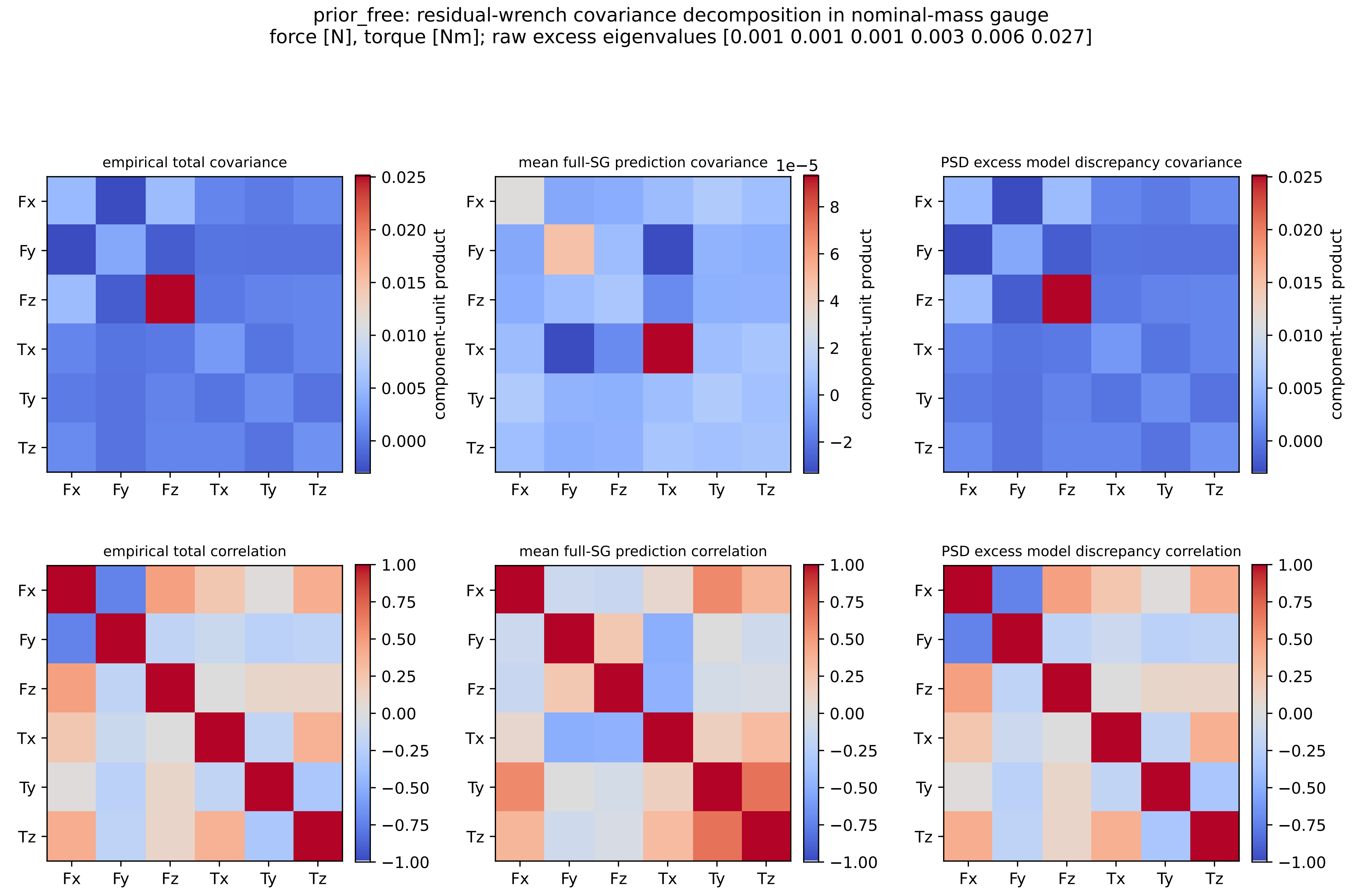
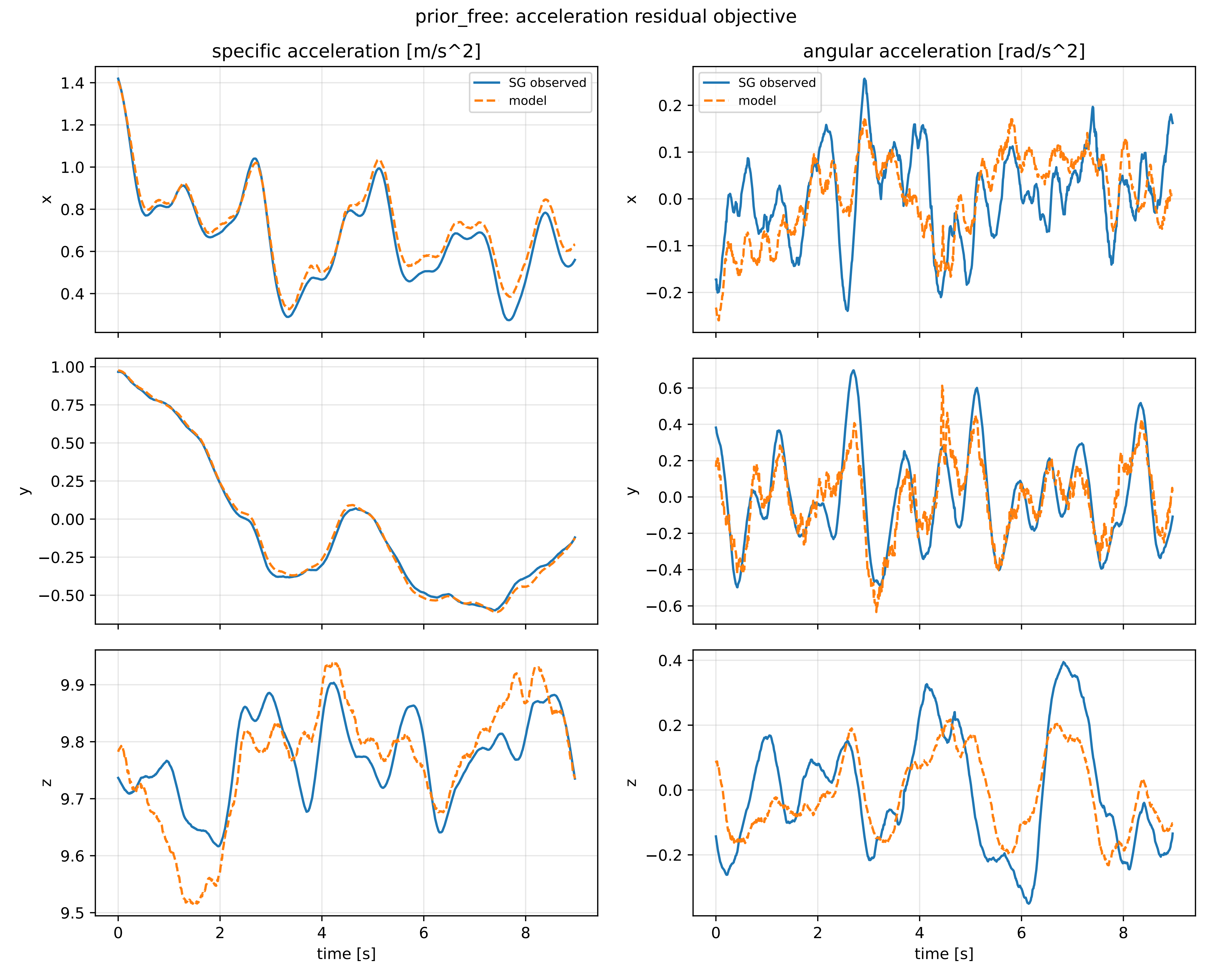
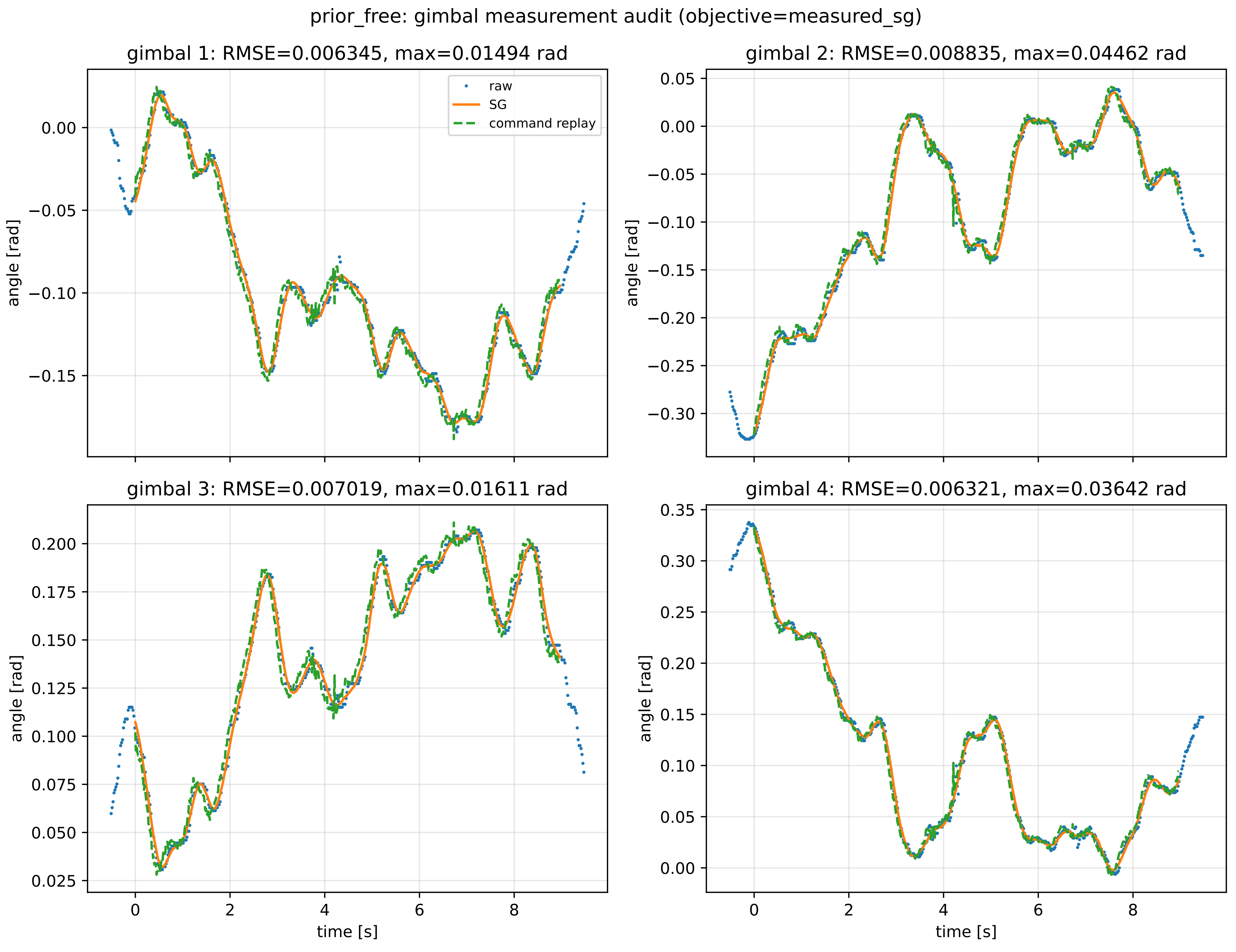

prior_free: SG trajectory and derived motion
pose position [m]
1.00
0.75
0.50
0.25
0.00
−0.25
x
y
z
body omega [rad/s]
0.15
0.10
0.05
0.00
−0.05
−0.10
−0.15
body alpha [rad/s^2]
0.6
0.4
0.2
0.0
−0.2
−0.4
0
2
4
6
8
time [s]



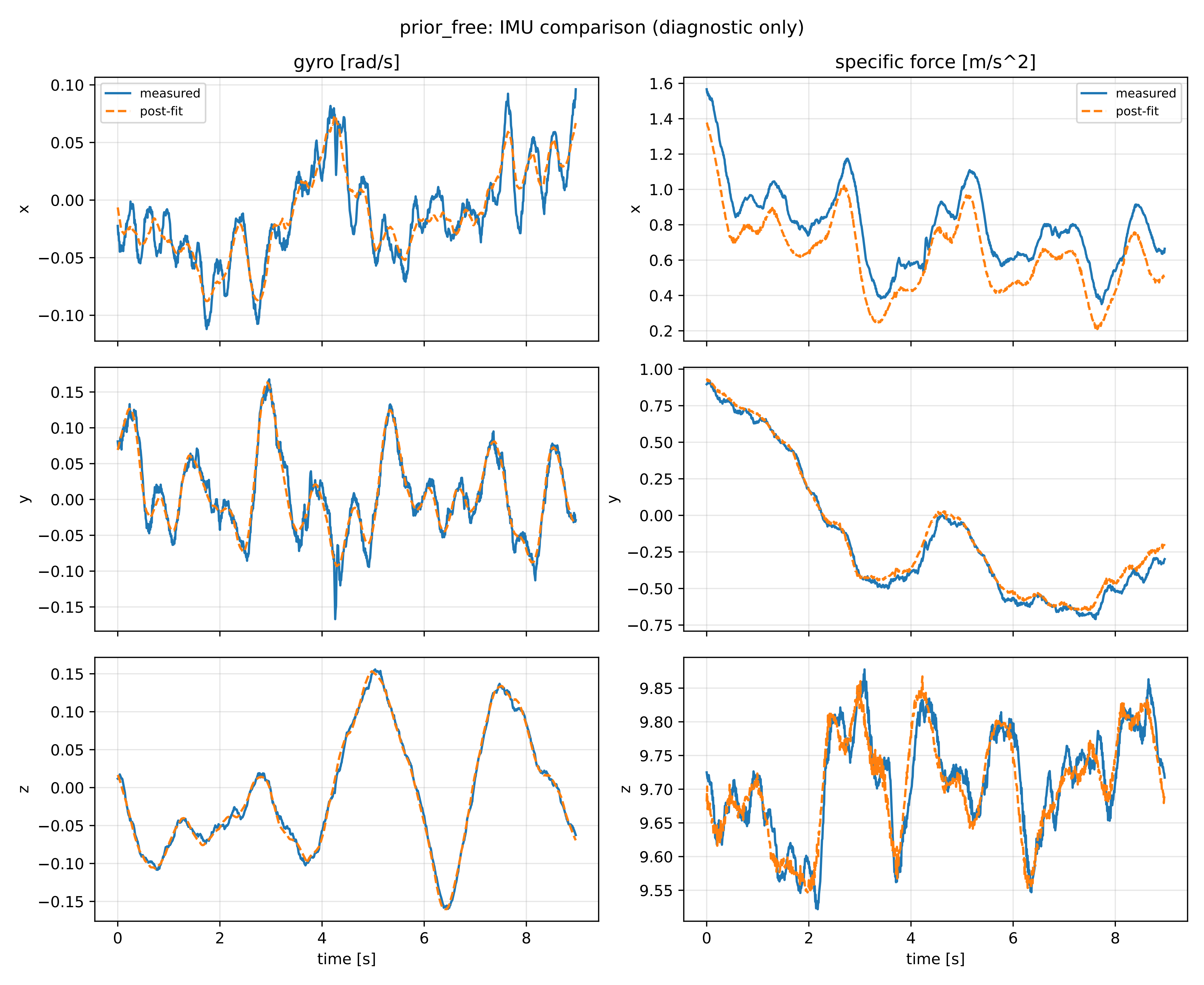
vector RMS about zero: force 0.20897 N, torque 0.072924 Nm; component std about mean: force [0.0709 0.0586 0.1585], torque [0.0478 0.0372 0.0406]

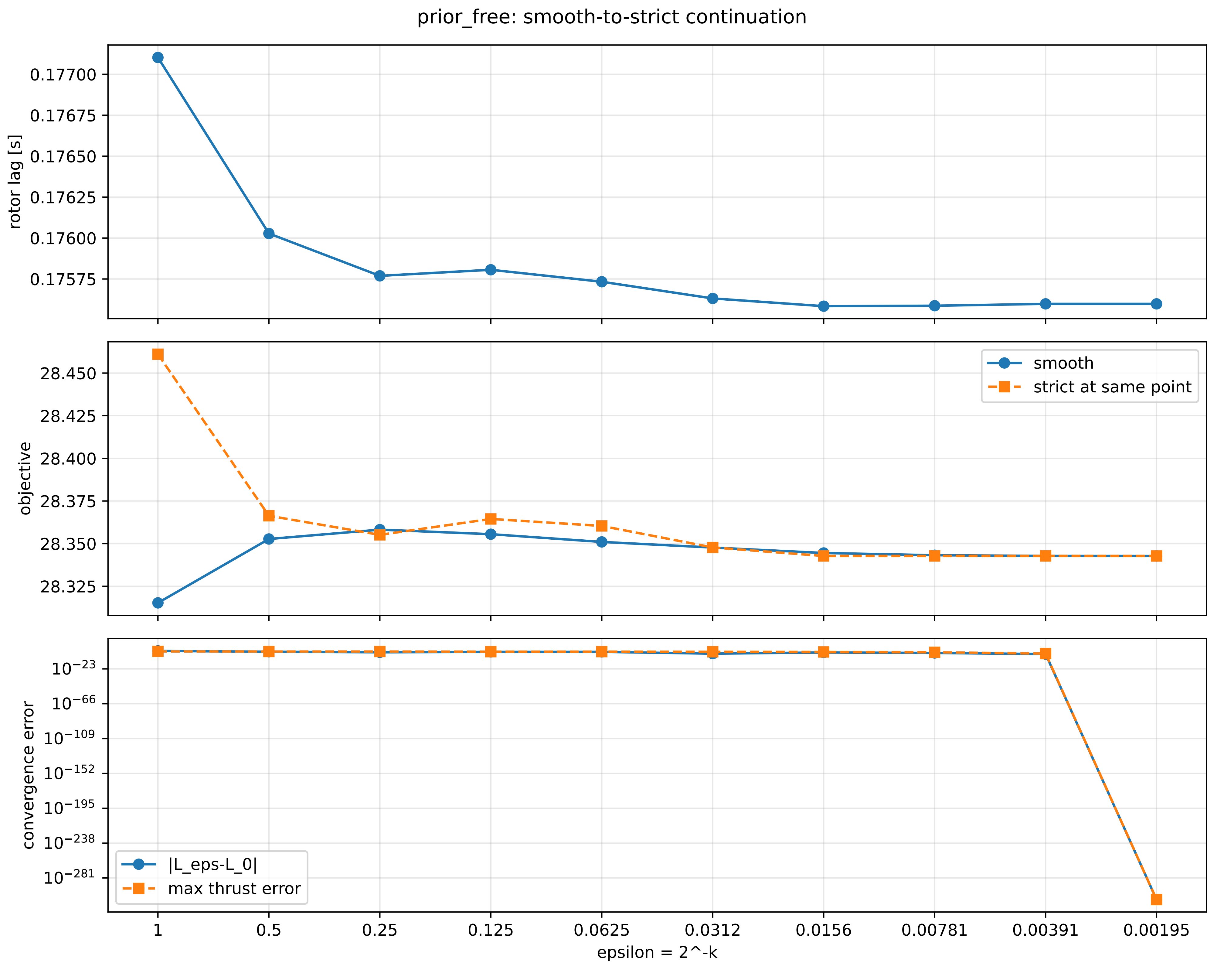

prior_free: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [0.001 0.001 0.001 0.003 0.006 0.027]
empirical total covariance
mean full-SG prediction covariance
PSD excess model discrepancy covariance
empirical total correlation
mean full-SG prediction correlation
PSD excess model discrepancy correlation
component-unit product
1e−5
Fx
Fy
Fz
Tx
Ty
Tz


prior_free: acceleration residual objective
specific acceleration [m/s^2]
angular acceleration [rad/s^2]
SG observed
model
x
y
z
time [s]

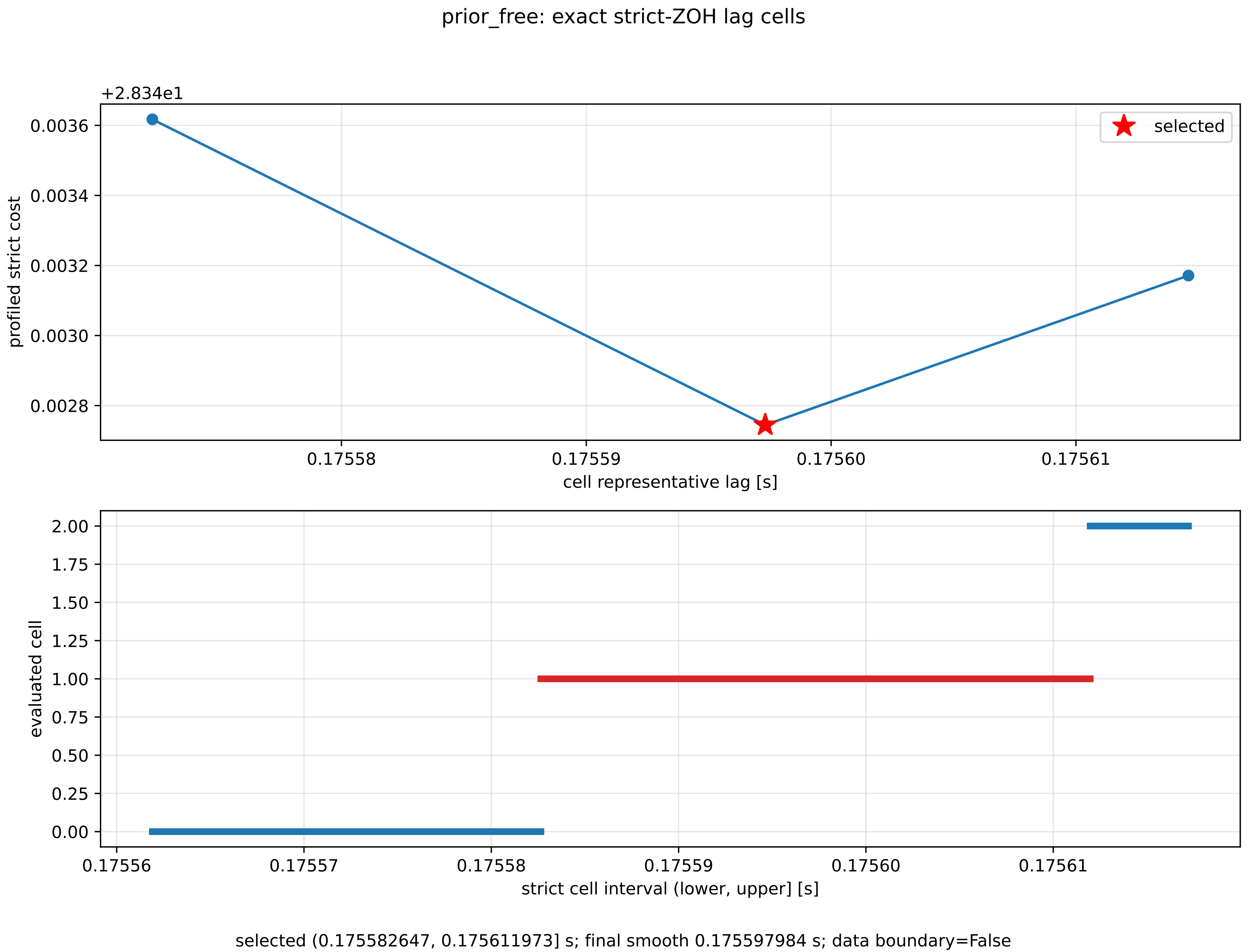


prior_free: gimbal measurement audit (objective=measured_sg)
gimbal 1: RMSE=0.006345, max=0.01494 rad
gimbal 2: RMSE=0.008835, max=0.04462 rad
gimbal 3: RMSE=0.007019, max=0.01611 rad
gimbal 4: RMSE=0.006321, max=0.03642 rad
raw
SG
command replay
angle [rad]
time [s]


prior_free: IMU comparison (diagnostic only)
gyro [rad/s]
specific force [m/s^2]
measured
post-fit
x
y
z
time [s]



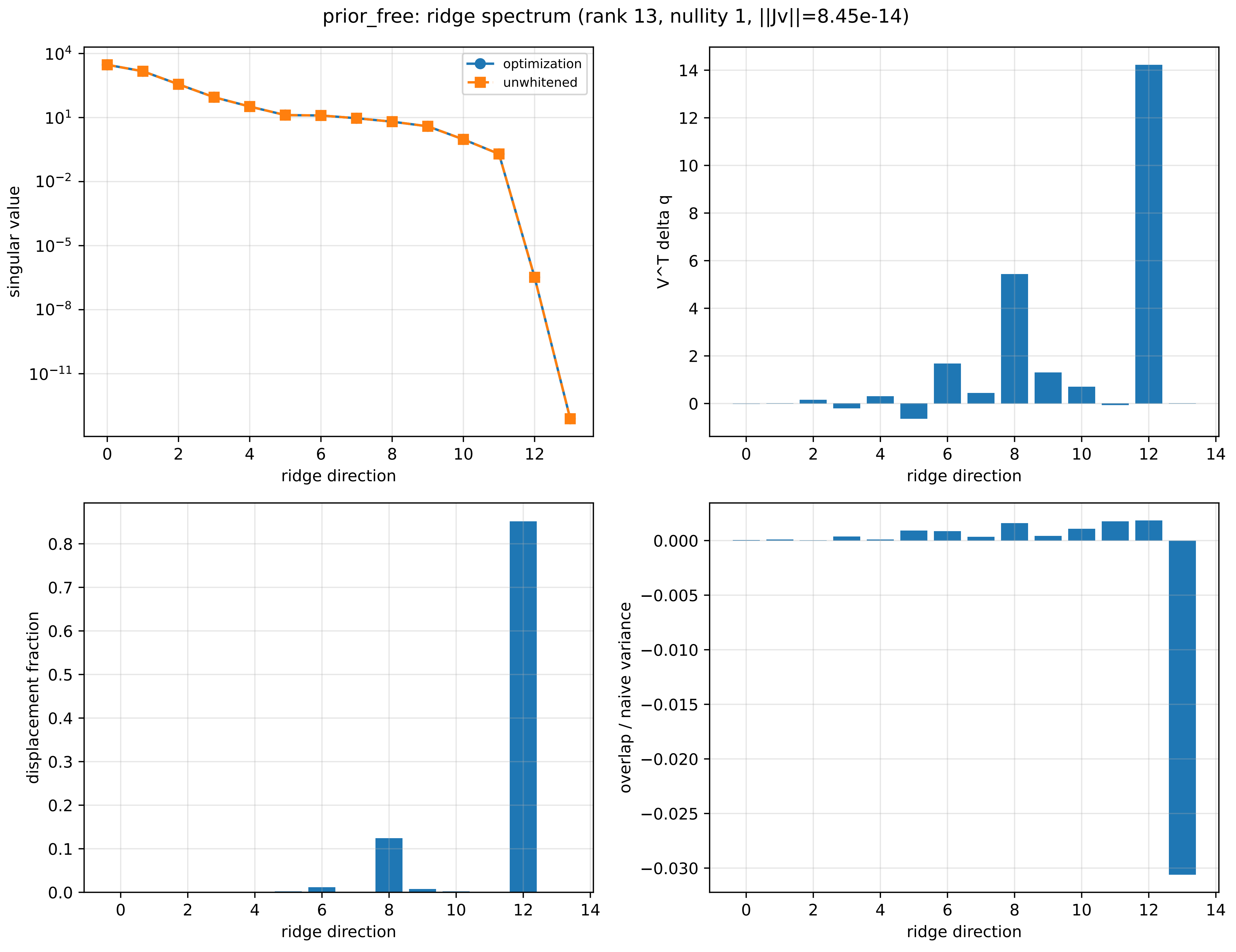
prior_free: smooth-to-strict continuation
rotor lag [s]
0.17700
0.17675
0.17650
0.17625
0.17600
0.17575
objective
28.450
28.425
28.400
28.375
28.350
28.325
smooth
strict at same point
convergence error
10^{-23}
10^{-66}
10^{-109}
10^{-152}
10^{-195}
10^{-238}
10^{-281}
|L_eps-L_0|
max thrust error
1
0.5
0.25
0.125
0.0625
0.0312
0.0156
0.00781
0.00391
0.00195
epsilon = 2^-k



prior_free: exact strict-ZOH lag cells
+2.834e1
0.0036
0.0034
0.0032
0.0030
0.0028
profiled strict cost
selected
0.17558
0.17559
0.17560
0.17561
cell representative lag [s]
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
evaluated cell
0.17556
0.17557
0.17558
0.17559
0.17560
0.17561
strict cell interval (lower, upper] [s]
selected (0.175582647, 0.175611973] s; final smooth 0.175597984 s; data boundary=False

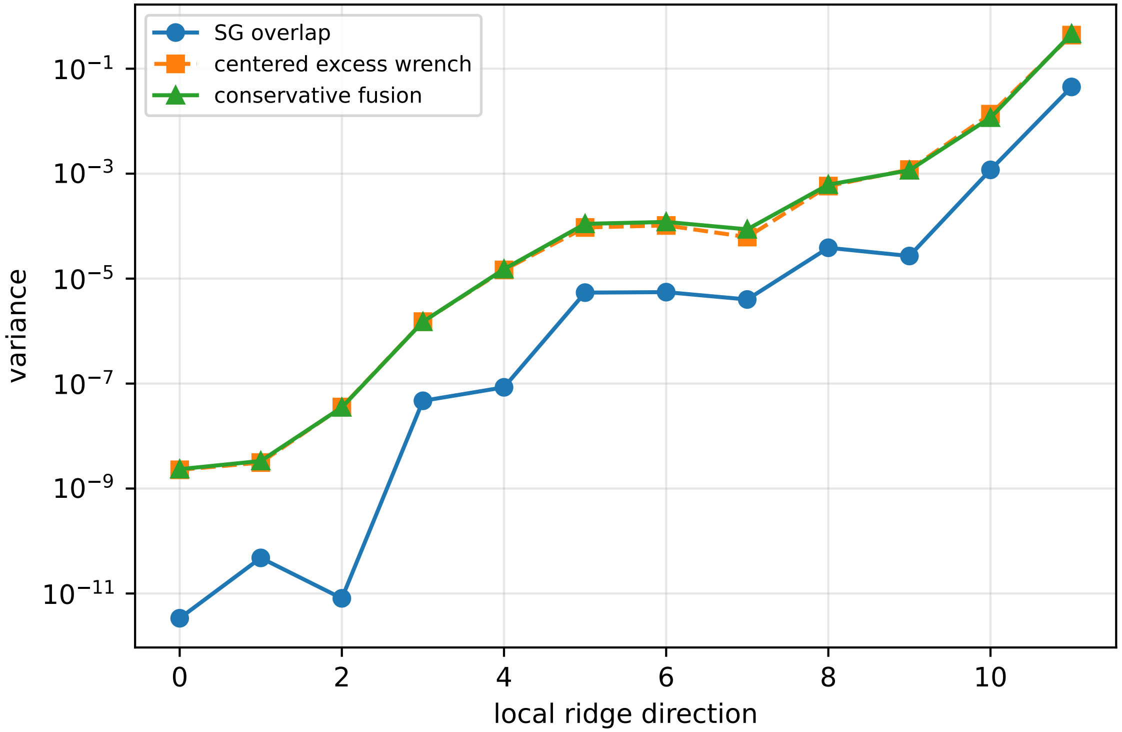
prior_free: parameter estimate

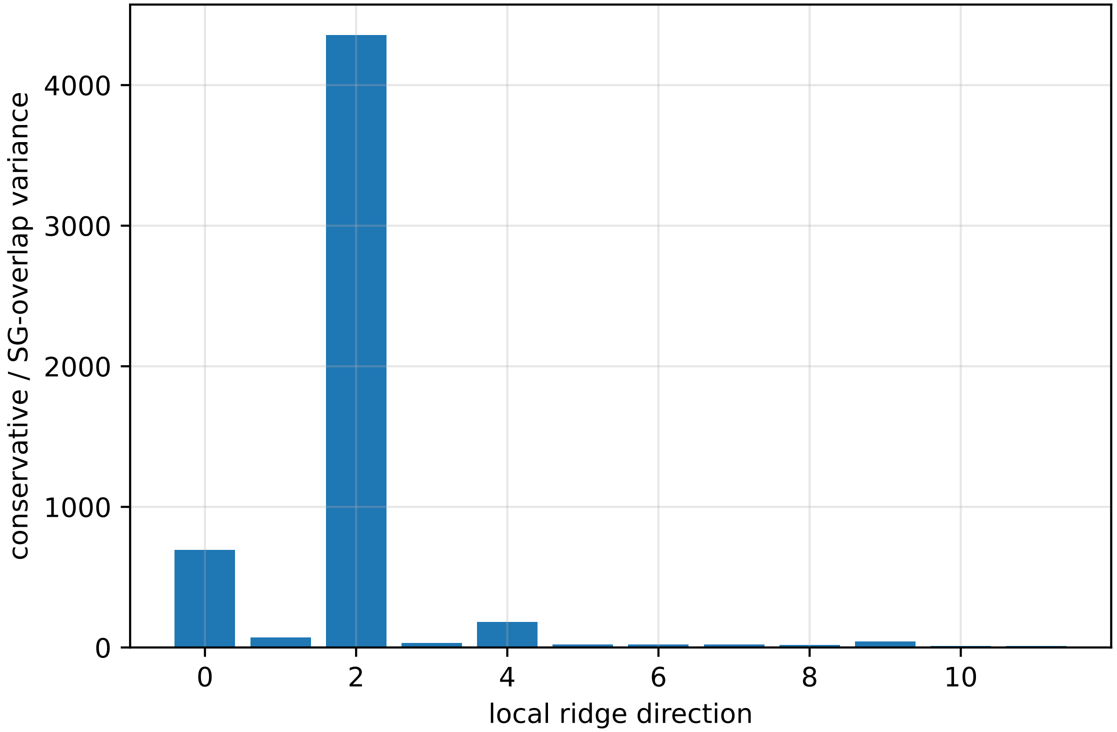
```
Case: prior_free
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 5.95395088
inertia [kg m^2]:
[[ 1.38088924 -0.03064693  0.10745121]
 [-0.03064693  0.65296997 -0.01821479]
 [ 0.10745121 -0.01821479  0.7674336 ]]
CoG body [m]: [-0.01982262 -0.05513055 -0.00503056]
force effectiveness: [2.6467129  2.58999676 1.36647131 1.65924695]

scale-free J/m [m^2]:
[[ 0.23192822 -0.00514733  0.01804704]
 [-0.00514733  0.10967003 -0.00305928]
 [ 0.01804704 -0.00305928  0.12889485]]
scale-free f/m: [0.44453052 0.43500472 0.22950665 0.27867999]

rotor lag cell: (0.175582647, 0.175611973] s
representative: 0.17559731 s
```

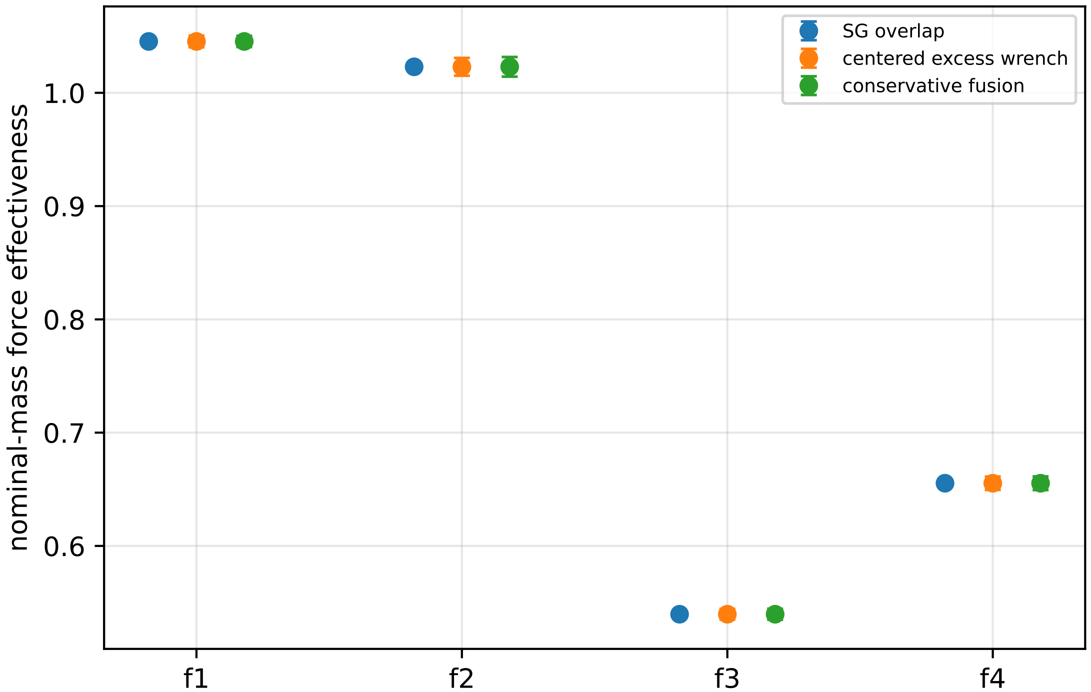


prior_free: ridge spectrum (rank 13, nullity 1, ||Jv||=8.45e-14)
singular value
ridge direction
optimization
unwhitened
V^T delta q
displacement fraction
overlap / naive variance

prior_free: Conservative fusion uncertainty



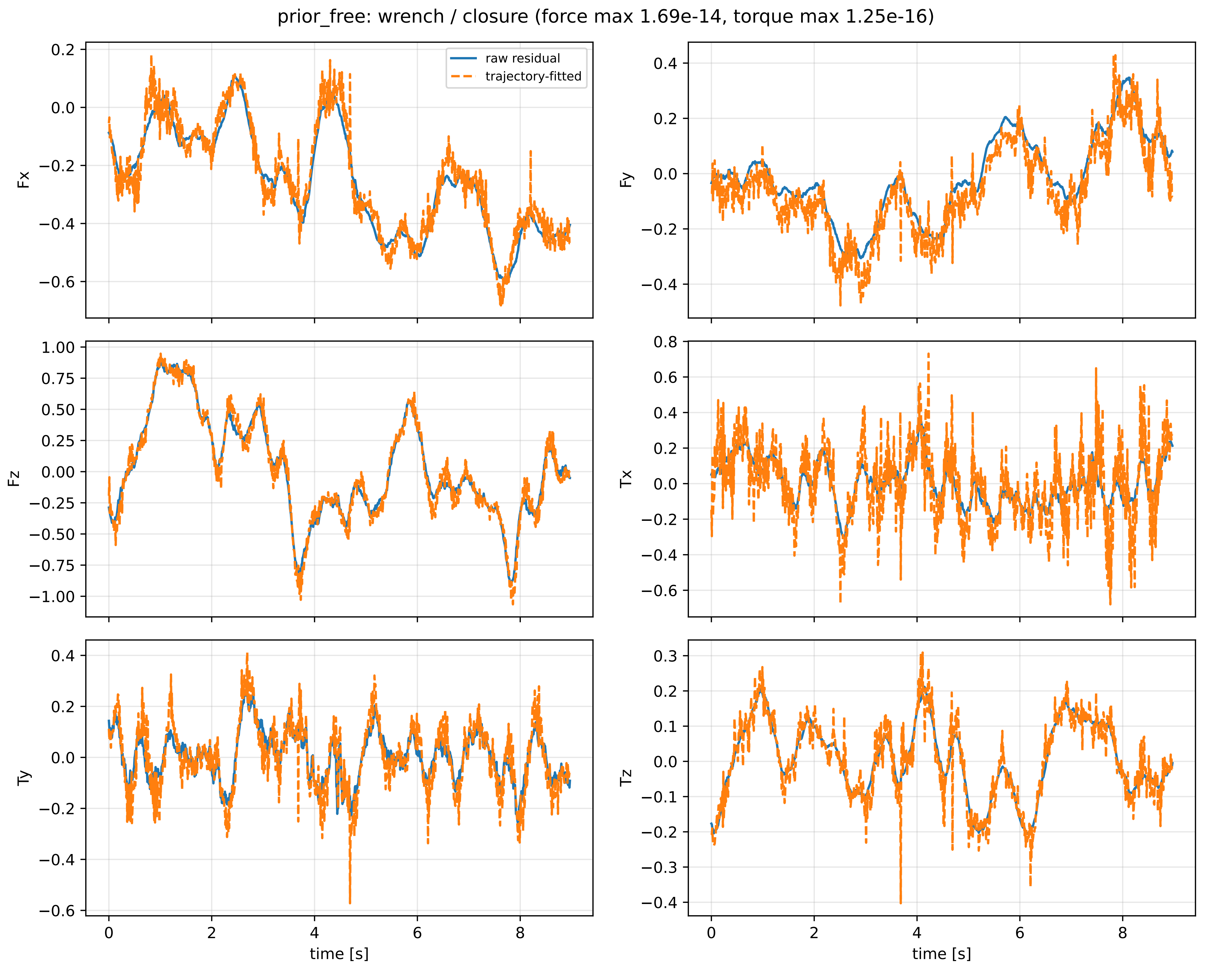





```
Top three non-gauge ambiguous directions
tr(M_res) / tr(M_SG) = 401.4
wrench-to-acceleration recovery max error = 2.86e-15
exact scale gauge ridge index = 13 (alignment 1)

index 11: variance=0.4624, ratio=10.3
  log_mass=+0.00157; second_moment_diag_1=+0.0653
  second_moment_diag_2=-0.0608; second_moment_diag_3=-0.0128
  second_moment_offdiag_12=+0.992; second_moment_offdiag_13=-0.0882
  second_moment_offdiag_23=+0.0236; cog_x=-0.000556
  cog_y=+0.000269; cog_z=-0.000373
  log_force_eff_1=+0.00331; log_force_eff_2=-0.00243
  log_force_eff_3=+0.0015; log_force_eff_4=+0.00438

index 10: variance=0.01168, ratio=9.901
  log_mass=-0.0185; second_moment_diag_1=-0.0144
  second_moment_diag_2=-0.0444; second_moment_diag_3=+0.142
  second_moment_offdiag_12=+0.0887; second_moment_offdiag_13=+0.981
  second_moment_offdiag_23=-0.0427; cog_x=+0.00582
  cog_y=-0.000797; cog_z=-0.00063
  log_force_eff_1=-0.0349; log_force_eff_2=+0.0056
  log_force_eff_3=+0.0195; log_force_eff_4=-0.0546

index 9: variance=0.00116, ratio=43.03
  log_mass=-0.169; second_moment_diag_1=+0.00103
  second_moment_diag_2=+0.893; second_moment_diag_3=-0.0375
  second_moment_offdiag_12=+0.0539; second_moment_offdiag_13=+0.0387
  second_moment_offdiag_23=+0.216; cog_x=-0.00428
  cog_y=-0.000447; cog_z=-0.0111
  log_force_eff_1=-0.164; log_force_eff_2=-0.172
  log_force_eff_3=-0.212; log_force_eff_4=-0.141
```

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

prior_free: optimization vs reference covariance
full Sigma_z eigenvalue
optimization whitening gain
Mahalanobis contribution
optimization metric
reference full covariance
time [s]


prior_free: wrench / closure (force max 1.69e-14, torque max 1.25e-16)
raw residual
trajectory-fitted
Fx
Fy
Fz
Tx
Ty
Tz
time [s]
time [s]